

Detection of Personal Protective Equipment Using Semi-Supervised Learning with YOLOv9-c

Amanda Aliyah^{1*}, Aisha Reissani Sopyan², Bryan Valleryan Alvonso³, Kenny Valent Winalda Sembiring⁴, & Vaurolla Az Zahra⁵

¹Department of Informatics, Multimedia Nusantara University, Tangerang, Banten, Indonesia

*Correspondence: amanda.aliyah@student.umn.ac.id

ABSTRACT

Personal Protective Equipment (PPE) detection is critical for worker safety on construction sites, yet deep learning models typically require large annotated datasets that are costly to obtain. We propose a semi-supervised learning (SSL) pipeline using YOLOv9-c, incorporating Programmable Gradient Information (PGI) and Generalized Efficient Layer Aggregation Network (GELAN), to exploit unlabelled data and reduce annotation overhead. The SH17 benchmark was merged with seven supplementary public PPE datasets, producing a 22,033-image, 12-class corpus with a 60/40 labelled–unlabelled split. Phase 1 established a supervised baseline (mAP@0.5 = 0.8308, Precision = 0.8596, Recall = 0.7995). Phase 2 applied adaptive pseudo-label self-training with per-class confidence thresholds calibrated from Phase 1 mAP@0.5, improving all four global metrics (mAP@0.5 +0.0060, mAP@0.5:0.95 +0.0132, Precision +0.0053, Recall +0.0127). The largest per-class gains occurred in body-protection (+0.038), face-mask (+0.023), and glasses (+0.010). EigenCAM visualisations confirmed accurate model attention on PPE regions across all 12 classes. An Intersection over PPE Area (IoPPE) metric is introduced to associate detections with individual workers, enabling three-state compliance assessment (Fully Compliant, Partially Compliant, Non-Compliant). The model achieves 38.4 FPS on an NVIDIA Tesla T4 GPU, demonstrating practical deployability for real-time safety monitoring under annotation-constrained conditions.

Keywords: *Personal protective equipment; Semi-supervised learning; YOLOv9; Object detection; Pseudo-labelling; Construction site safety; Explainable AI*

Introduction

Occupational health and safety (OHS) represents a persistent global challenge in the construction and manufacturing sectors, which consistently record the highest rates of fatal and non-fatal workplace injuries worldwide. According to the International Labour Organization, the construction industry alone accounts for more than 30% of fatal occupational accidents globally despite employing only 7–10% of the workforce¹. Mandatory use of Personal Protective Equipment (PPE) including safety helmets, high-visibility vests, protective gloves, safety footwear, and face protection is a regulatory cornerstone in most national OHS frameworks; however, manual compliance monitoring by on-site supervisors is operationally impractical at scale, prone to human error, and incapable of providing continuous real-time coverage across large worksites. Advances in deep learning-based computer vision have created compelling opportunities for automating PPE compliance monitoring. The YOLO (You Only Look Once) family of single-stage detectors has been widely adopted in this domain owing to its favourable balance of accuracy and real-time inference speed, with applications spanning helmet and mask detection^{2,3}, multi-class PPE monitoring in industrial and construction environments^{4,7,19,20}, and edge-deployed compliance systems^{5,6,8}. Among recent variants, YOLOv9-c⁹ introduces Programmable Gradient Information (PGI), which preserves informative gradient signals through an auxiliary reversible branch during training, mitigating the information bottleneck inherent in deep networks. PGI is coupled with the Generalized Efficient Layer Aggregation Network (GELAN), which provides computationally efficient multi-scale feature aggregation using conventional convolution operators. Compared to YOLOv5 and YOLOv7, YOLOv9-c achieves superior mAP on MS-COCO while maintaining competitive inference throughput, making it well-suited for deployment on surveillance hardware with limited computational resources.

A persistent challenge in deploying PPE detection models is the dependency on large quantities of manually annotated training data. Unlabelled surveillance footage is routinely generated on construction sites but remains unused in conventional fully-supervised pipelines. Semi-supervised learning (SSL) addresses this limitation by enabling models to learn from both a small labelled set and a large unlabelled corpus through pseudo-labelling and self-training. Prior SSL methods for general object detection include STAC¹⁰, which applies data augmentation consistency to pseudo-labels; Unbiased Teacher¹¹, which

uses a Mean Teacher framework with class-rebalancing focal loss; and SoftTeacher¹², which exploits soft pseudo-label weighting. Recent work has also demonstrated SSL utility in construction site monitoring^{13,14}. However, these approaches have not been systematically evaluated on multi-class PPE detection benchmarks with more than six PPE categories, and none has been applied to the comprehensive SH17 dataset. Existing PPE detection work has predominantly relied on fully-supervised training, limiting practical deployability under annotation-constrained conditions. To address this, we propose an SSL framework where the 60/40 labelled-to-unlabelled split is specifically designed to simulate real-world data scarcity; this configuration balances the requirement for a robust initial supervised baseline with the necessity of maintaining a sufficient unlabelled corpus for effective self-training. This paper addresses these research gaps through four principal contributions. First, we present a complete adaptive pseudo-label SSL pipeline using YOLOv9-c with per-class confidence thresholds calibrated from Phase 1 mAP@0.5 to mitigate class imbalance. Second, we assemble an augmented 12-class PPE corpus of 22,033 images from eight public datasets, with targeted oversampling of minor classes. Third, we provide quantitative demonstration that SSL achieves mAP@0.5 of 0.8368 using only 60% labelled data, improving over the supervised baseline across all four evaluation metrics. Fourth, we contribute EigenCAM visual explanations and introduce an Intersection over PPE Area (IoPPE) metric that extends raw detection outputs to actionable per-worker compliance assessment.

Related Work

Computer vision-based PPE detection has progressed rapidly, with YOLO-family detectors emerging as the dominant approach due to their real-time inference capability. Fully-supervised studies have demonstrated strong performance across varying class cardinalities and deployment contexts: Wang et al.¹⁵ achieved 86.55% mAP@0.5 using YOLOv5x on a three-class construction dataset; Kumar et al.¹⁶ reported 76.86% mAP with YOLOv4 across six PPE categories; Lo et al.¹⁷ attained 97.0% mAP with YOLOv7 on a four-class custom dataset; and Wang et al.²⁸ established a new state-of-the-art using YOLOv10 with Vision Transformer on the full 17-class SH17 benchmark. Further fully-supervised work has addressed helmet and mask detection^{2,3}, edge-deployed lightweight systems^{5,7,8}, small object detection under complex industrial conditions^{4,27}, compliance monitoring for workers at height and hazardous environments²⁵, head-mounted and body-worn PPE categories^{18,19,20,22}, and multi-class benchmarking on standardised datasets^{23,24,26}. Despite strong performance, these methods share a reliance on complete annotation, limiting deployability where labelling budget is constrained. SSL addresses this by leveraging unlabelled data through pseudo-labelling and self-training. Key developments include STAC¹⁰, Unbiased Teacher¹¹, SoftTeacher¹², and the Efficient Teacher framework³⁰ designed specifically for one-stage anchor-based detectors; adaptive threshold methods such as LabelMatch³¹ have further improved pseudo-label quality under long-tailed distributions. Recent work has applied SSL to construction site monitoring^{13,14}, though none has evaluated it on a comprehensive multi-class PPE benchmark exceeding six categories or on the SH17 dataset. PPE compliance assessment beyond raw detection, including per-worker item assignment and partial compliance evaluation has been pursued through pose-guided spatial anchoring³², pose estimation-based worker-PPE matching³³, and multi-camera worker re-identification combined with PPE classification³⁴. For model explainability, EigenCAM³⁵ provides a gradient-free spatial attribution map suitable for multi-anchor detectors such as YOLO, where Grad-CAM is impractical due to the absence of a single scalar class score; its utility in object detection explainability has been demonstrated in drone imagery and surveillance contexts³⁶. The present study is the first, to our knowledge, to combine YOLOv9-c, adaptive SSL pseudo-labelling, and EigenCAM explainability on a 12-class PPE corpus.

Methods

Dataset Collection and Curation

The primary dataset is SH17 (Safety Helmet 17)²⁹, comprising 8,099 images with YOLO-format bounding box annotations across 17 original classes collected from diverse industrial environments. Following class analysis, 12 PPE-relevant classes were retained: person (0), face (1), face-mask (2), foot (3), glasses (4), gloves (5), helmet (6), hands (7), head (8), body-protection (9, merged from medical-suit and safety-suit), shoes (10), and safety-vest (11). Classes ear, ear-muffs, face-guard, and tool were excluded as non-PPE or peripherally relevant to worksite safety compliance. Seven supplementary public datasets were integrated to address annotation scarcity in minor classes: (i) a helmet-only Roboflow dataset; (ii) a combined glasses/gloves/helmet/safety-boots/safety-vest dataset; (iii) the aiotthien PPE dataset (17 classes, remapped); (iv) PPE-kelompok4 (gloves, glasses, face-mask, shoes); (v) the CPPE-5 Medical PPE dataset (COCO JSON, converted to

YOLO format); and (vi–vii) two foot-detection datasets. The final corpus comprised 22,033 images with 104,929 annotations.

Data Preprocessing and Splitting

Preprocessing comprised four sequential steps: (1) format conversion from COCO to YOLO, applied to the CPPE-5 dataset to standardise annotations; (2) cross-dataset class mapping, remapping labels from the original 17-class SH17 schema to a unified 12-class PPE taxonomy, including merging medical-suit and safety-suit into body-protection; (3) dataset merging via symbolic links, consolidating images from all eight source datasets into a single centralised directory; and (4) exploratory data analysis (EDA) to quantify class distributions and guide training strategy selection. To address the extreme class imbalance identified during EDA, the Copy-Paste augmentation technique was applied specifically to minor classes (safety-vest, body-protection) to synthesize additional instances against varied backgrounds. Finally, the merged corpus of 22,033 images was partitioned using a fixed random seed (seed = 42), producing a 60% labelled subset (~13,171 images) for Phase 1 supervised training and a 40% unlabelled subset (~8,862 images) for pseudo-label generation in Phase 2, with a held-out validation set of 4,775 images retained for monitoring and early stopping.

Model Architecture

YOLOv9-c was selected as the detection backbone for three principal reasons⁹. First, PGI preserves informative gradient signals via an auxiliary reversible branch during training, mitigating information loss in deep networks that arises from the information bottleneck principle. Second, GELAN enables computationally efficient multi-scale feature aggregation using conventional convolution operators, achieving state-of-the-art parameter utilisation on MS-COCO without relying on depth-wise separable convolutions. Third, YOLOv9-c achieves competitive mAP on MS-COCO relative to prior YOLO variants (YOLOv5, YOLOv7, YOLOv8), providing a robust initialisation for domain transfer. The model was initialised from COCO-pretrained weights with the output head modified to predict 12 PPE-specific classes. The SPPELAN module at backbone layer 9 (model.model[9]) produces a $\sim 20 \times 20$ feature map with 512 channels, representing the richest semantic abstractions prior to the detection neck.

Phase 1: Supervised Baseline Training

Phase 1 trained YOLOv9-c on the 60% labelled split for 25 epochs using Stochastic Gradient Descent (SGD) with cosine learning rate decay (initial lr = 0.01, momentum = 0.937, weight decay = 0.0005). Input resolution was fixed at 640×640 pixels. Aggressive data augmentation was applied: mosaic (p = 1.0), copy-paste (p = 0.8), MixUp (p = 0.25), and HSV colour jitter. The classification loss weight was set to 2.0 to emphasise class discrimination. Training used dual NVIDIA Tesla T4 GPUs with Distributed Data Parallel (DDP), batch size 16 per GPU, and global seed 42. Mosaic augmentation was disabled in the final 10 epochs (close_mosaic = 10) to stabilise convergence. Three warmup epochs were applied at the start of training.

Phase 2: Adaptive Pseudo-Label Generation and SSL Self-Training

Following Phase 1 convergence, the best checkpoint (mAP@0.5 = 0.8308) was applied in inference mode to all 8,862 unlabelled images. Per-class adaptive confidence thresholds were set proportional to Phase 1 mAP@0.5 for each class: high-performing classes (helmet, mAP = 0.977) received strict thresholds (0.65), while low-performing classes (shoes, mAP = 0.651) received lenient thresholds (0.28) to maximise pseudo-label recall for underrepresented categories without introducing excessive noise in high-confidence classes. NMS IoU threshold was set to 0.45. This procedure yielded pseudo-labels for 7,883 images (88.9% acceptance rate) with 33,233 new bounding box annotations, forming a combined SSL training set of 22,033 images. Phase 2 fine-tuned the Phase 1 checkpoint for 20 epochs at a reduced learning rate (lr = 0.001). Backbone layers 0–9 were frozen to preserve feature representations learned during Phase 1. Classification loss weight was reduced to 1.5, while Copy-Paste and MixUp augmentation probabilities were reduced to 0.3 and 0.10, respectively. Mosaic augmentation remained enabled with close_mosaic = 5, and two warmup epochs were applied at the start of training. Early stopping monitored validation mAP@0.5 with a patience of 15 epochs.

Evaluation Metrics

Model performance was quantified using: mAP@0.5 (mean Average Precision at IoU threshold 0.5), mAP@0.5:0.95 (averaged across IoU thresholds 0.5–0.95 in steps of 0.05), Precision (positive predictive value), and Recall (true positive

rate). All metrics were computed on the held-out validation set using the Ultralytics evaluation engine. Per-class AP@0.5 was computed separately for each of the 12 PPE categories. Inference speed was benchmarked as FPS on a single NVIDIA Tesla T4 GPU.

EigenCAM Explainability Analysis

EigenCAM³⁵ was applied to the SPPELAN module at backbone layer 9, producing a $\sim 20 \times 20$ feature map with 512 channels representing the richest semantic abstractions prior to the detection neck. EigenCAM was preferred over Grad-CAM because the YOLO detection head generates thousands of anchor-level predictions simultaneously, making a single scalar class score unavailable for gradient computation. EigenCAM computes the principal eigenvector of the feature map via Singular Value Decomposition (SVD) and uses it as a class-agnostic spatial attribution map without requiring a backward pass. Activation maps were generated for representative validation images from each of the 12 PPE classes to qualitatively verify that the model had learned semantically meaningful representations. Per-class confidence distributions were computed on a 200-image validation sample to assess adaptive threshold calibration quality.

PPE Compliance Assessment: Intersection over PPE Area (IoPPE)

Prior approaches to worker–PPE spatial association have relied on pose estimation keypoints^{32,33} or on worker re-identification across multiple camera feeds³⁴. To associate PPE detections with individual worker detections without requiring auxiliary models, we define the Intersection over PPE Area (IoPPE) as the ratio of the intersection area between a PPE bounding box and a person bounding box to the PPE bounding box area alone:

$$\text{IoPPE} = \text{Area}(\text{PPE} \cap \text{Person}) / \text{Area}(\text{PPE})$$

This metric corrects the geometric asymmetry inherent in standard IoU: a helmet occupies approximately 5% of a person bounding box area, so the maximum achievable IoU is approximately 0.05 even when the helmet is correctly worn. IoPPE produces values approaching 1.0 for any fully contained PPE item regardless of the size ratio between the item and the worker bounding box. Mandatory PPE items for classification as Fully Compliant were helmet (class ID 6) and safety-vest (class ID 11). Optional items recorded but not penalised in compliance scoring included face-mask (2), glasses (4), gloves (5), and shoes (10). Each detected worker was assigned one of three compliance states: Fully Compliant (both mandatory items detected above the IoPPE threshold, visualised in green); Partially Compliant (exactly one mandatory item detected, visualised in orange); or Non-Compliant (no mandatory items detected, visualised in red).

Use of AI Assistance

Large Language Models (LLMs) were used to assist with manuscript editing and English-language proofreading. No LLM contributed to the experimental design, data collection, analysis, or interpretation of results, and no LLM is listed as an author. Use of AI assistance is declared in accordance with Scientific Reports editorial policy.

Results

Dataset Composition and Class Distribution

The augmented training corpus yielded 22,033 images and 104,929 annotations across 12 PPE-relevant classes (Table 1). Exploratory data analysis revealed pronounced class imbalance: major classes such as hands (15,850 annotations) and person (13,802) vastly outnumber minor classes such as safety-vest (530), foot (759), and body-protection (927), directly informing the adaptive threshold calibration and oversampling strategies described in the Methods section. The maximum imbalance ratio was $11.7\times$ (hands vs. body-protection), comparable to long-tailed distributions reported in other multi-class safety datasets.

ID	Class Name	Annotations	Tier	Phase 1 mAP@0.5
0	person	13,802	Major	0.896
1	face	8,950	Major	0.900
2	face-mask	804	Minor	0.778
3	foot	759	Minor	0.888

4	glasses	1,945	Mid	0.736
5	gloves	2,790	Mid	0.681
6	helmet	927	Minor	0.977
7	hands	15,850	Major	0.861
8	head	11,985	Major	0.898
9	body-protection	927	Minor	0.820
10	shoes	4,560	Mid	0.651
11	safety-vest	530	Minor	0.882

Table 1. Final 12-class distribution of the augmented PPE corpus after merging SH17 with seven supplementary datasets.

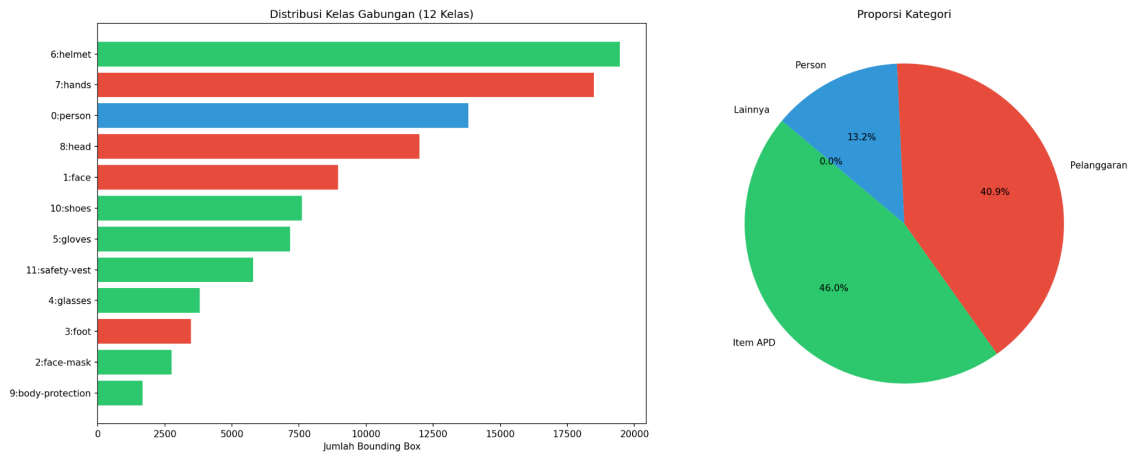


Figure 1. Per-class annotation count distribution across the merged 12-class PPE corpus.

Phase 1: Supervised Baseline Training

Phase 1 trained YOLOv9-c on 13,171 labelled images (60% of the corpus) for 25 epochs on dual NVIDIA Tesla T4 GPUs using DDP. The global results are summarised in Table 2. Per-class mAP@0.5 ranged from 0.651 (shoes) to 0.977 (helmet); major classes (person, face, head) exceeded 0.89, while minor and mid-tier classes were lower (gloves: 0.681, glasses: 0.736, face-mask: 0.778). The Phase 1 normalised confusion matrix showed dominant diagonal values (>0.80) for eight of the twelve classes, indicating reliable class discrimination with limited inter-class confusion.

Parameter	Phase 1 Value	Phase 2 Value
Epochs	25	20
Image Size	640 × 640	640 × 640
Batch Size	16 per GPU (×2 T4)	16 per GPU (×2 T4)
Optimizer	SGD (cosine LR)	SGD (cosine LR)
Learning Rate	0.01	0.001
Momentum	0.937	0.937

Weight Decay	0.0005	0.0005
Mosaic	1.0	1.0
Copy-Paste	0.8	0.3
Mixup	0.25	0.10
Close Mosaic	10	5
cls loss weight	2.0	1.5
Warmup Epochs	3	2
Backbone Frozen Layers	None	0–9

Table 2. Training hyperparameters used in Phase 1 (supervised baseline) and Phase 2 (SSL self-training).

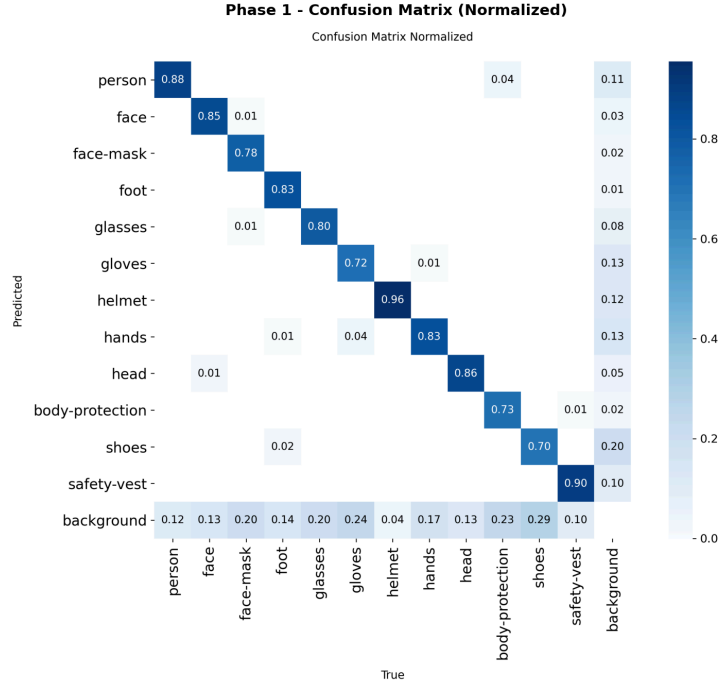


Figure 2. Normalised confusion matrix for Phase 1 (supervised baseline).

Phase 2: Pseudo-Label Generation and SSL Self-Training

The Phase 1 model was applied to 8,862 unlabelled images, yielding pseudo-labels for 7,883 images (88.9% acceptance rate) containing 33,233 new bounding box annotations. Per-class adaptive confidence thresholds (Table 3) were calibrated from Phase 1 mAP@0.5: high-performing classes received stricter thresholds (helmet: 0.65, person: 0.60) while low-performing classes received more lenient thresholds (shoes: 0.28, gloves: 0.30) to maximise pseudo-label recall for underrepresented instances. Backbone layers 0–9 were frozen during Phase 2 to prevent pseudo-label noise from degrading Phase 1 feature representations. Training curves showed stable convergence without substantial overfitting throughout the 20 training epochs.

ID	Class	Phase 1 mAP@0.5	Conf. Threshold	Rationale
0	person	0.896	0.60	High mAP: strict threshold applied
1	face	0.900	0.58	High mAP: strict threshold applied

2	face-mask	0.778	0.30	Lower mAP: lenient threshold to recover instances
3	foot	0.888	0.40	Moderate mAP: moderate threshold
4	glasses	0.736	0.35	Lower mAP: lenient threshold
5	gloves	0.681	0.30	Lower mAP: lenient threshold
6	helmet	0.977	0.65	Highest mAP: very strict threshold
7	hands	0.861	0.52	Moderate mAP: moderate threshold
8	head	0.898	0.58	High mAP: strict threshold applied
9	body-protection	0.820	0.35	Lower mAP: lenient threshold
10	shoes	0.651	0.28	Lowest mAP: most lenient threshold
11	safety-vest	0.882	0.52	Moderate mAP: moderate threshold

Table 3. Adaptive per-class confidence thresholds used during pseudo-label generation.

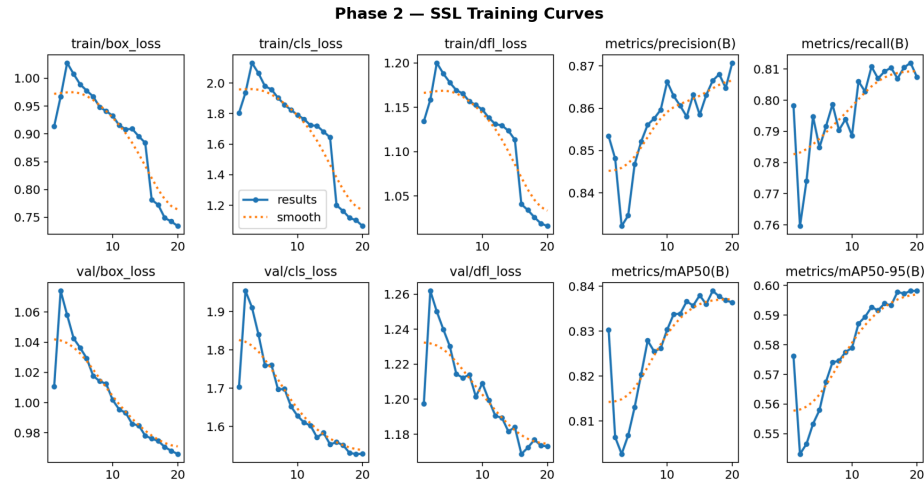


Figure 3. Phase 2 training curves showing box loss, classification loss, DFL loss, $mAP@0.5$, and $mAP@0.5:0.95$ convergence over 20 epochs.

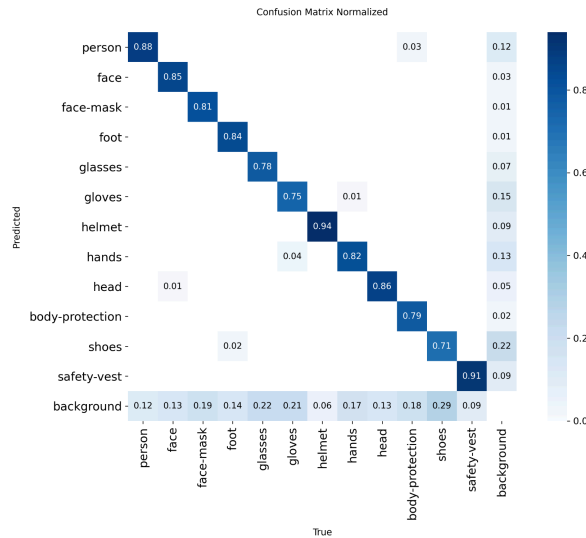


Figure 4. Normalised confusion matrix for Phase 2 (SSL).

Metric	Phase 1 (60% labeled)	Phase 2 (SSL)	Δ (absolute)
mAP@0.5	0.8308	0.8368	+0.0060
mAP@0.5:0.95	0.5849	0.5981	+0.0132
Precision	0.8596	0.8649	+0.0053
Recall	0.7995	0.8122	+0.0127

Table 4. Global evaluation metrics for Phase 1 (supervised baseline) and Phase 2 (SSL).

Global and Per-Class Performance Comparison

SSL training (Phase 2) consistently improved all four global evaluation metrics over the Phase 1 supervised baseline (Table 4). The largest absolute gain was in mAP@0.5:0.95 (+0.0132), indicating improved bounding box localisation precision across stricter IoU thresholds beyond IoU = 0.5. The improvement in Recall (+0.0127) indicates that Phase 2 recovers additional true positive detections that the Phase 1 model missed. The improvement in Precision (+0.0053) confirms that the adaptive threshold filtering strategy maintained pseudo-label quality and did not introduce substantial false positive noise. Real-time inference speed was 38.4 FPS on a single NVIDIA Tesla T4 GPU, satisfying requirements for live CCTV-based safety monitoring.

Class	Tier	Phase 1	Phase 2	Δ	Trend
person	Major	0.896	0.901	+0.005	↑
face	Major	0.900	0.902	+0.002	↑
face-mask	Minor	0.778	0.801	+0.023	↑
foot	Minor	0.888	0.895	+0.007	↑
glasses	Mid	0.736	0.746	+0.010	↑
gloves	Mid	0.681	0.688	+0.007	↑
helmet	Minor	0.977	0.974	−0.003	↓
hands	Major	0.861	0.860	−0.001	→
head	Major	0.898	0.901	+0.003	↑
body-protection	Minor	0.820	0.858	+0.038	↑
shoes	Mid	0.651	0.650	−0.001	→
safety-vest	Minor	0.882	0.868	−0.014	↓

Table 5. Per-class mAP@0.5 for Phase 1 and Phase 2.

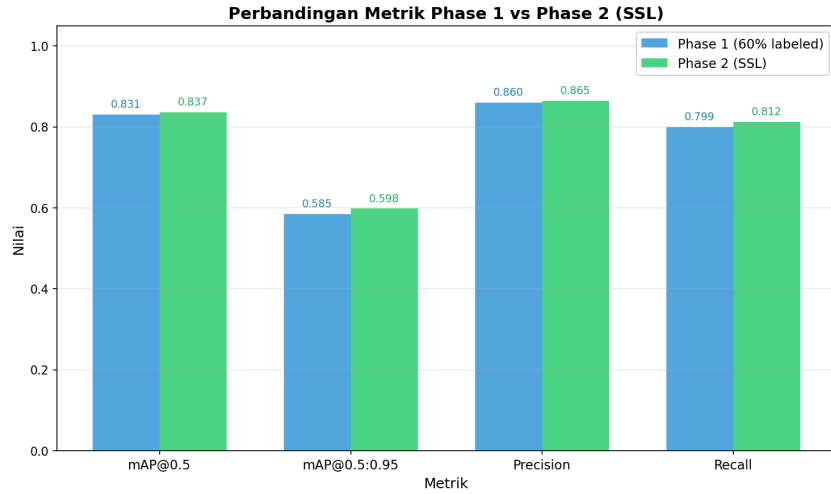


Figure 5. Bar chart comparison of overall detection metrics between Phase 1 (60% labeled) and Phase 2 (SSL).

Per-class analysis (Table 5) shows 8 of 12 classes improved, 1 remained stable (hands: -0.001), and 3 showed marginal degradation (helmet: -0.003 , safety-vest: -0.014 , shoes: -0.001). The largest gains were in body-protection ($+0.038$), face-mask ($+0.023$), and glasses ($+0.010$), all minor or mid-tier classes, confirming that lenient per-class thresholds effectively directed pseudo-label coverage towards underrepresented categories.

Comparison with Prior Fully-Supervised Approaches

Table 6 compares the proposed SSL pipeline with representative fully-supervised approaches. The proposed method achieves 83.68% mAP@0.5 using only 60% labelled data. Direct comparison with prior studies should be interpreted cautiously because reported performance depends strongly on dataset composition, class definitions, annotation quality, and evaluation protocols. Unlike many previous studies that focus on two to six PPE categories, the proposed framework addresses 12 PPE-related classes while reducing annotation requirements through semi-supervised learning. These results demonstrate that competitive detection performance can be achieved without relying exclusively on fully labelled datasets.

Study	Model	Dataset	Labeled	mAP@0.5	SSL
Wang et al. (2021)	YOLOv5x	CHV (1,330 imgs)	100%	86.55%	No
Lo et al. (2022)	YOLOv7	Custom (11K images)	100%	97.0%	No
Ferdous & Ahsan (2022)	Custom YOLO	CHVG (4 cls)	100%	~85%	No
Lee et al. (2023)	YOLO+Tracking	Industrial PPE dataset	100%	66.4%	No
Kumar et al. (2021)	YOLOv4	14,500 images	100%	76.86%	No
Wang et al. (2025)	YOLOv10+ViT	SH17 (17 cls)	100%	87.32%	No
This Study	YOLOv9-c+SSL	SH17+8 sources (12 cls)	60%	83.68%	Yes

Table 6. Comparison of the proposed SSL pipeline with prior fully-supervised PPE detection methods.

XAI: EigenCAM Visualisation and Confidence Analysis

EigenCAM was applied to the SPPELAN module at backbone layer 9. Activation maps for representative validation images across all 12 classes showed consistent localisation to semantically correct PPE regions: helmet surfaces, vest regions, glove contours, facial protection areas, and foot-level shoe detections, with minimal background activation. This suggests that the model has learned task-relevant representations rather than relying primarily on confounding image artefacts.

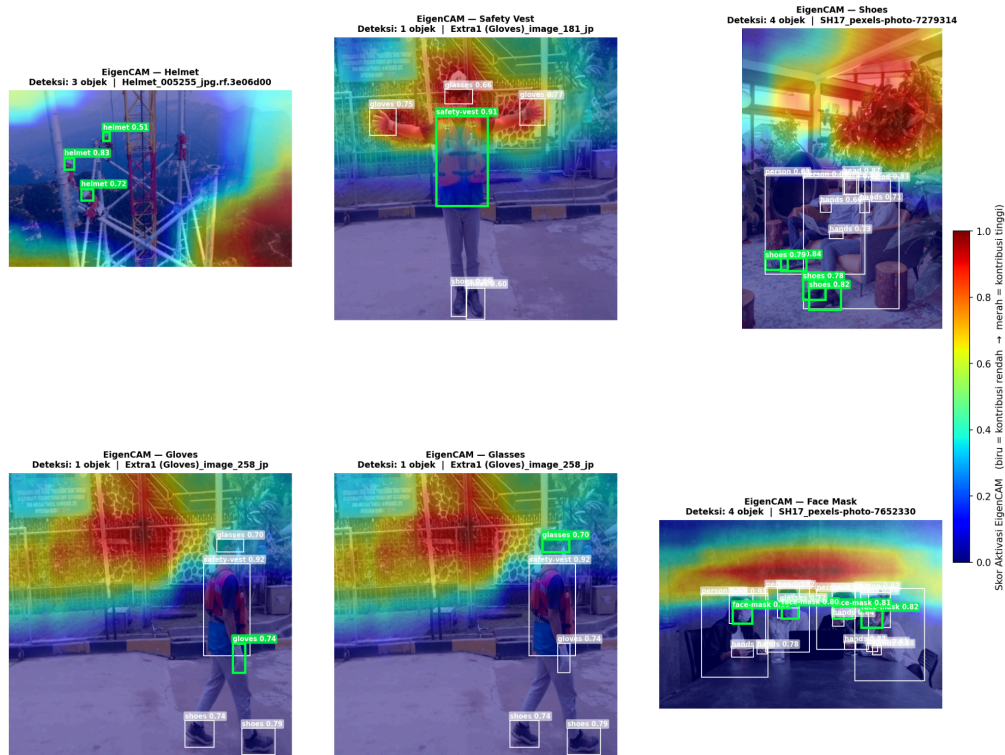


Figure 6. EigenCAM activation maps for representative PPE detection examples across 12 classes.

Per-class confidence distributions on a 200-image validation sample confirmed that 10 of 12 class thresholds were well-calibrated, with mean confidence clearly exceeding the adaptive threshold. Class shoes and gloves showed mean confidence close to their thresholds, indicating these classes remain candidates for threshold refinement or supplementary annotation in subsequent SSL iterations. The confidence histograms for high-performing classes (helmet, face) showed strongly right-skewed distributions with most mass above 0.7, while minor-class histograms were more dispersed, consistent with their lower per-class mAP.

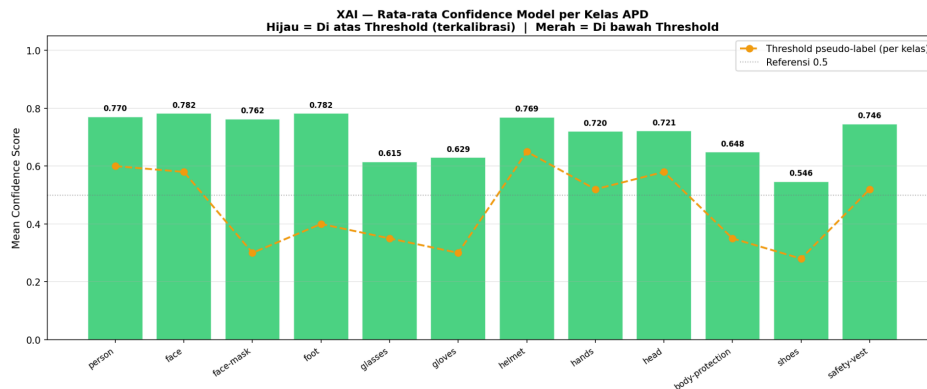


Figure 7. Mean confidence score per class (bars) versus adaptive pseudo-label threshold (orange dashed line).

XAI – Distribusi Confidence Score per Kelas APD
Merah putus-putus = Mean Model | Oranye titik-titik = Threshold Pseudo-label

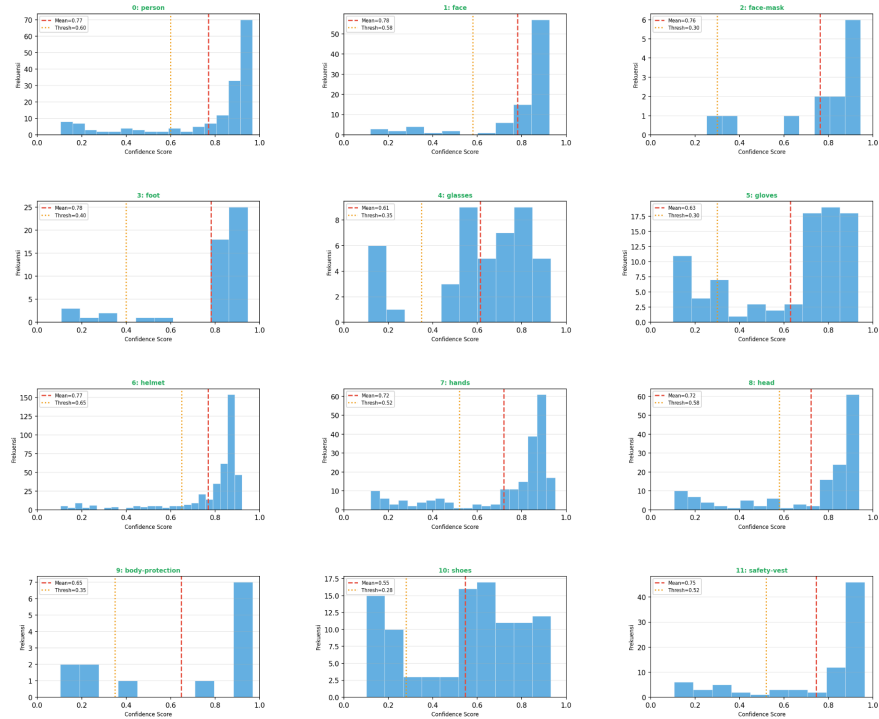


Figure 8. Histograms of per-class confidence score distributions on a 200-image validation sample.

PPE Compliance Assessment via Intersection over PPE Area (IoPPE)

The IoPPE-based compliance state machine was applied to representative validation images. Results showed that the system correctly identified workers wearing both mandatory items (helmet and safety-vest) as Fully Compliant (green bounding box), workers with only one mandatory item as Partially Compliant (orange), and workers with no mandatory items as Non-Compliant (red). Non-construction contexts (e.g., office workers, individuals photographed outdoors) were consistently flagged as Non-Compliant, demonstrating that the compliance logic correctly depends on active detection rather than position or appearance heuristics. Inference at 38.4 FPS supports live video deployment on standard GPU-equipped monitoring hardware.

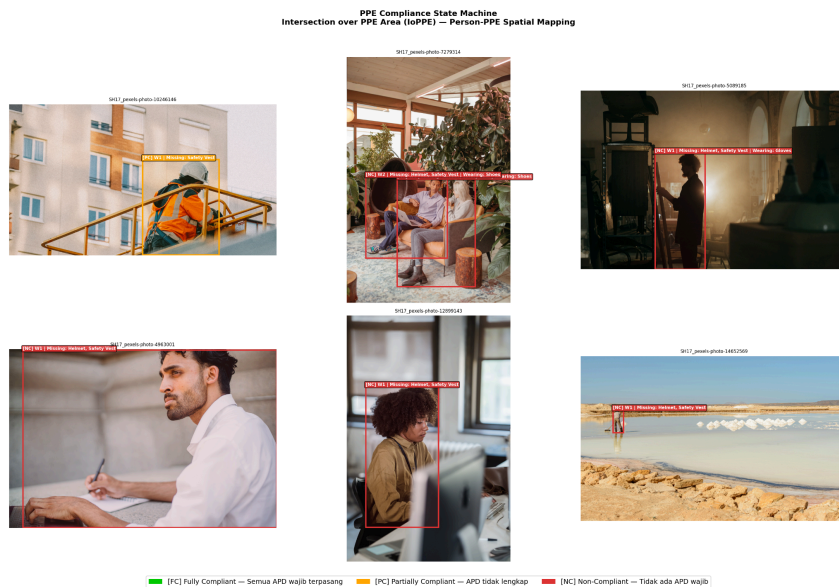


Figure 9. Compliance state machine output showing per-worker compliance status across representative validation images.

Discussion

The SSL pipeline demonstrates that exploiting 40% unlabelled data through adaptive pseudo-label self-training consistently improves all global detection metrics over a supervised baseline trained on 60% of the labelled corpus. The 60/40 labelled-to-unlabelled ratio was selected to represent a realistic annotation-constrained scenario while retaining sufficient labelled data for a robust supervised baseline. Although the observed gains are modest in absolute terms (mAP@0.5 +0.0060, mAP@0.5:0.95 +0.0132), this outcome is consistent with prior semi-supervised object detection studies^{10,11,12} operating on strong supervised baselines. When baseline performance already exceeds 0.83 mAP@0.5, substantial improvements become increasingly difficult to obtain. The consistent improvement across all four evaluation metrics suggests that the additional pseudo-labelled data provided useful supervisory signals without noticeably degrading model performance.

The most substantive per-class gains occurred in minor and mid-tier categories, specifically body-protection (+0.038), face-mask (+0.023), and glasses (+0.010), precisely those assigned the most lenient adaptive thresholds (0.30–0.35). This observation supports the effectiveness of class-adaptive threshold calibration under long-tailed class distributions, consistent with findings reported by LabelMatch31 and Efficient Teacher30. Marginal degradation in helmet (−0.003) likely reflects near-ceiling baseline performance (0.977) rather than SSL failure. In contrast, the decrease in safety-vest performance (−0.014) highlights a bottleneck in the current pipeline; error analysis reveals this is primarily driven by severe annotation scarcity, combined with motion blur in high-activity zones and heavy occlusion when workers carry tools or materials. These factors—compounded by the class’s inherent visual variability—remain the most significant challenges for robust PPE detection in uncontrolled environments. Future SSL iterations may benefit from targeted data collection and specific threshold refinement for this class.

Direct comparison with prior fully supervised studies should be interpreted cautiously because reported mAP values depend strongly on dataset composition, class definitions, annotation quality, and evaluation protocols. Unlike many previous studies that focus on a smaller number of PPE categories, the proposed framework addresses 12 PPE-related classes while reducing annotation requirements through semi-supervised learning. EigenCAM visualisations localised consistently to semantically meaningful PPE regions across all classes, providing qualitative evidence that the model learned task-relevant representations. In addition, the proposed IoPPE metric was designed to address the geometric limitations of conventional IoU when associating small PPE items with worker detections and provides a practical mechanism for three-state compliance assessment. Limitations of this study include the use of a single SSL iteration, a fixed 60/40 labelled-to-unlabelled split, and evaluation on curated public datasets rather than live CCTV footage. Future work should investigate iterative multi-round SSL, alternative split ratios, active learning strategies, comparison against a fully supervised YOLOv9-c model trained on 100% labelled data, and deployment in real-world construction environments.

Conclusion

This study presented a semi-supervised learning pipeline for multi-class PPE detection combining the YOLOv9-c architecture with adaptive pseudo-label self-training on a merged 22,033-image corpus spanning 12 PPE-relevant classes. Phase 2 SSL achieved mAP@0.5 of 0.8368, mAP@0.5:0.95 of 0.5981, Precision of 0.8649, and Recall of 0.8122 using only 60% labelled data, consistently improving over the supervised baseline across all four metrics. Adaptive per-class confidence thresholds calibrated to Phase 1 mAP directed pseudo-label coverage toward underrepresented categories, yielding the largest gains in body-protection (+0.038), face-mask (+0.023), and glasses (+0.010). EigenCAM visual explanations confirmed semantically accurate model attention across all 12 PPE classes, providing additional qualitative evidence for model interpretability. The IoPPE compliance state machine translates raw detections into per-worker Fully Compliant, Partially Compliant, and Non-Compliant status, demonstrating practical deployability for automated safety monitoring at 38.4 FPS. The findings suggest that semi-supervised learning can reduce annotation requirements while maintaining competitive detection performance, potentially lowering the cost and effort associated with developing PPE monitoring systems for industrial environments. This pipeline represents a meaningful step toward annotation-efficient, interpretable PPE compliance monitoring systems suitable for real-world construction and manufacturing site deployment.

References

1. International Labour Organization. Safety and health at work: A vision for sustainable prevention. ILO (2023). <https://www.ilo.org/resource/news/nearly-3-million-people-die-work-related-accidents-and-diseases>
2. Li, S., Lv, Y., Liu, X. et al. Detection of safety helmet and mask wearing using improved YOLOv5s. Scientific Reports 13, 21417 (2023). <https://doi.org/10.1038/s41598-023-48943-3>
3. Fu, L. Construction site safety helmet wearing detection method based on improved YOLOv5. Journal of Physics: Conference Series 2560, 012042 (2023). <https://doi.org/10.1088/1742-6596/2560/1/012042>
4. Li, X., Hu, M., Li, B. & Tong, R. OAM-YOLO: A real-time small object detection framework for PPE compliance monitoring in industrial environments. Process Safety and Environmental Protection 194, 108058 (2025). <https://doi.org/10.1016/j.psep.2025.108058>
5. Tahiramani, N. et al. Edge-based AI solution for enhancing urban safety: helmet compliance monitoring with YOLOv9 on Raspberry Pi. Discover Internet of Things 5, 25 (2025). <https://doi.org/10.1007/s43926-025-00113-9>
6. Nazli, N. A. N. M. et al. A real-time system for detecting personal protective equipment compliance using deep learning model YOLOv5. Procedia Computer Science 252, 3494–3503 (2024). <https://doi.org/10.1016/j.procs.2024.10.291>
7. Ke, X., Chen, W. & Guo, W. 100+ FPS detector of personal protective equipment for worker safety: A deep learning approach for green edge computing. Peer-to-Peer Networking and Applications 15, 950–972 (2022). <https://doi.org/10.1007/s12083-021-01258-4>
8. Aminuddin, N. et al. Real-time personal protective equipment compliance detection using You Only Look Once. In Proceedings of the 2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS), 1–6 (IEEE, 2024). <https://doi.org/10.1109/AiDAS63860.2024.10730560>
9. Wang, C.-Y., Yeh, I.-H. & Liao, H.-Y. M. YOLOv9: Learning what you want to learn using programmable gradient information. In Computer Vision – ECCV 2024, Lecture Notes in Computer Science 15089, 1–21 (Springer, Cham, 2025). https://doi.org/10.1007/978-3-031-72751-1_1
10. Sohn, K., Zhang, Z., Li, C.-L., Zhang, H., Lee, C.-Y. & Pfister, T. A simple semi-supervised learning framework for object detection. arXiv preprint arXiv:2005.04757 (2020). <https://doi.org/10.48550/arXiv.2005.04757>
11. Liu, Y.-C., Ma, C.-Y., He, Z. et al. Unbiased teacher for semi-supervised object detection. arXiv preprint arXiv:2102.09480 (2021). <https://doi.org/10.48550/arXiv.2102.09480>
12. Xu, M., Zhang, Z., Hu, H. et al. End-to-end semi-supervised object detection with soft teacher. arXiv preprint arXiv:2106.09018 (2021). <https://doi.org/10.48550/arXiv.2106.09018>
13. Xiao, B., Zhang, Y., Chen, Y. & Yin, X. A semi-supervised learning detection method for vision-based monitoring of construction sites by integrating teacher-student networks and data augmentation. Advanced Engineering Informatics 50, 101413 (2021). <https://doi.org/10.1016/j.aei.2021.101413>
14. Kim, H.-S., Seong, J. & Jung, H.-J. Leveraging semisupervised learning for domain adaptation: Enhancing safety at construction sites through long-tailed object detection. Journal of Construction Engineering and Management 151, 1 (2025). <https://doi.org/10.1061/JCEMD4.COENG-15259>
15. Wang, Z., Wu, Y., Yang, L., Thirunavukarasu, A., Evison, C. & Zhao, Y. Fast personal protective equipment detection for real construction sites using deep learning approaches. Sensors 21, 3478 (2021). <https://doi.org/10.3390/s21103478>
16. Kumar, S. et al. YOLOv4 algorithm for the real-time detection of fire and personal protective equipments at construction sites. Multimedia Tools and Applications 81, 22163–22183 (2022). <https://doi.org/10.1007/s11042-021-11280-6>
17. Lo, C.-C., Lin, C.-H. & Hung, M.-H. Real-time personal protective equipment compliance detection based on deep learning algorithm. Sustainability 14, 16591 (2022). <https://doi.org/10.3390/su142416591>
18. Hayat, A. & Morgado-Dias, F. Deep learning-based automatic safety helmet detection system for construction safety. Applied Sciences 12, 8268 (2022). <https://doi.org/10.3390/app12168268>
19. Gallo, G. et al. A smart system for personal protective equipment detection in industrial environments based on deep learning at the edge. IEEE Access 10, 110862–110878 (2022). <https://doi.org/10.1109/ACCESS.2022.3215148>
20. Isailovic, V. et al. The compliance of head-mounted industrial PPE by using deep learning object detectors. Scientific Reports 12, 16347 (2022). <https://doi.org/10.1038/s41598-022-20282-9>

21. Lee, Y.-R. et al. Deep learning-based framework for monitoring wearing personal protective equipment on construction sites. *Journal of Computational Design and Engineering* 10, 905–917 (2023). <https://doi.org/10.1093/jcde/qwad019>
22. Ferdous, M. & Ahsan, S. M. M. PPE detector: A YOLO-based architecture to detect personal protective equipment for construction sites. *PeerJ Computer Science* 8, e999 (2022). <https://doi.org/10.7717/peerj-cs.999>
23. Barlybayev, A. et al. Personal protective equipment detection using YOLOv8 architecture on object detection benchmark datasets: a comparative study. *Cogent Engineering* 11, 2333209 (2024). <https://doi.org/10.1080/23311916.2024.2333209>
24. Liu, Y. & Wang, J. Personal protective equipment detection for construction workers: A novel dataset and enhanced YOLOv5 approach. *IEEE Access* 12, 47338–47358 (2024). <https://doi.org/10.1109/ACCESS.2024.3382817>
25. Chen, H. et al. YOLOv5s-gnConv: Detecting personal protective equipment for workers at height. *Frontiers in Public Health* 11, 1225478 (2023). <https://doi.org/10.3389/fpubh.2023.1225478>
26. Baoju, L., Xiangqian, W., Qingshan, C. et al. Safety helmet detection methods in heavy machinery factory. *Scientific Reports* 15, 18565 (2025). <https://doi.org/10.1038/s41598-025-02824-z>
27. Li, C. et al. SD-YOLOv5: A rapid detection method for personal protective equipment on construction sites. *Frontiers in Built Environment* 11, 1563483 (2025). <https://doi.org/10.3389/fbuil.2025.1563483>
28. Wang, S. et al. Automated non-PPE detection on construction sites using YOLOv10 and transformer architectures for surveillance and body worn cameras with benchmark datasets. *Scientific Reports* 15, 27043 (2025). <https://doi.org/10.1038/s41598-025-12468-8>
29. Ahmad, H. M. & Rahimi, A. SH17: A dataset for human safety and personal protective equipment detection in manufacturing industry. *Journal of Safety Science and Resilience* 6, 175–185 (2025). <https://doi.org/10.1016/j.jnlssr.2024.09.002>
30. Xu, B., Chen, M., Guan, W. & Hu, L. Efficient Teacher: Semi-Supervised Object Detection for YOLOv5. *arXiv preprint arXiv:2302.07577* (2023). <https://doi.org/10.48550/arXiv.2302.07577>
31. Chen, B., Chen, W., Yang, S., Xuan, Y., Song, J., Xie, D., Pu, S., Song, M. & Zhuang, Y. Label matching semi-supervised object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14381–14390 (2022). <https://arxiv.org/abs/2206.06608>
32. Xiong, R. & Tang, P. Pose guided anchoring for detecting proper use of personal protective equipment. *Automation in Construction* 130, 103828 (2021). <https://doi.org/10.1016/j.autcon.2021.103828>
33. López, L., Suárez-Ramírez, J., Alemán-Flores, M. & Monzón, N. Automated PPE compliance monitoring in industrial environments using deep learning-based detection and pose estimation. *Automation in Construction* (2025). <https://doi.org/10.1016/j.autcon.2025.105802>
34. Cheng, Z. et al. Vision-based monitoring of site safety compliance based on worker re-identification and personal protective equipment classification. *Automation in Construction* 139, 104312 (2022). <https://doi.org/10.1016/j.autcon.2022.104312>
35. Muhammad, M. B. & Yeasin, M. Eigen-CAM: Visual explanations for deep convolutional neural networks. *SN Computer Science* 2, 47 (2021). <https://doi.org/10.1007/s42979-021-00449-3>
36. Joshi, G. et al. Ensemble learning and EigenCAM-based feature analysis for improving the performance and explainability of object detection in drone imagery. *Discover Applied Sciences* (2025). <https://doi.org/10.1007/s42452-025-06879-5>

Author Contributions

K.V.W.S. conceived the research idea and designed the overall study framework. K.V.W.S., B.V.A., and V.A.Z. collected and curated the datasets used in this study. A.R.S. developed and refined the model training code, supported implementation, and prepared the project presentation materials. K.V.W.S., B.V.A., and V.A.Z. conducted model training, experimentation, and performance evaluation. A.A. performed the literature review, organized and interpreted the experimental findings, prepared figures and tables, and wrote the manuscript. All authors contributed to the discussion of results, reviewed the manuscript, and approved the final version.

Competing Interests

The authors declare no competing interests.

Data Availability

The SH17 primary dataset is publicly available at <https://github.com/ahmadmughees/SH17dataset>. Supplementary datasets are publicly available via Roboflow Universe and Kaggle. The merged dataset configuration, class mappings, and train-validation split indices used in this study will be made available through the project repository upon publication.

Code Availability

The code developed for data preprocessing, pseudo-label generation, adaptive threshold calibration, EigenCAM visualisation, and PPE compliance assessment is available from the corresponding author upon reasonable request. The YOLOv9 implementation is publicly available from the official repository: <https://github.com/WongKinYiu/yolov9>

Acknowledgements

The authors thank the creators of the SH17 dataset (H. M. Ahmad and A. Rahimi, University of Windsor) and the maintainers of the supplementary public PPE datasets used in this study. GPU computing resources were provided via Google Colaboratory. No external funding was received for this study.